

Supply Chain Performance Analytics for GlamourGlow Cosmetics

Project



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Imaginary Company: GlamourGlow Cosmetics

About the Company:

GlamourGlow Cosmetics is a fast-scaling beauty brand operating on a **hybrid B2C and B2B model**. The company produces and sells makeup products across both digital and retail channels. With a global supplier base and a multi-warehouse logistics network, the business operates at the intersection of high-demand, seasonal variability, and customer expectation for fast delivery.

As the company expands, its **supply chain complexity has escalated**—bringing inefficiencies and blind spots into focus. GlamourGlow has hired you as a **Data Analyst** to build a **comprehensive Tableau dashboard** that turns fragmented supply chain data into **strategic, operational, and real-time visibility** for decision-makers.

Problem Statement

Despite rapid growth, GlamourGlow faces **systemic inefficiencies in its supply chain**:

- **Inventory Imbalance:** Stockouts and overstocks coexist due to poor visibility into inventory health.
- **Supplier Volatility:** Lead times and defect rates vary dramatically across sourcing partners.
- **Fulfillment Lags:** While 95% of orders are on time, late deliveries are not well-tracked or addressed.
- **Cost Escalation:** Fluctuating fuel and freight charges impact profitability.
- **Data Siloing:** Operations rely on disconnected spreadsheets, causing strategic blind spots.

The leadership team needs a **data-driven command center**—a Tableau dashboard that centralizes KPIs, detects bottlenecks, and supports proactive planning.

Dataset Link: [Supply Chain Dataset](#)

Dataset Overview

The dataset includes integrated supply chain dimensions across manufacturing, warehousing, order fulfillment, and transportation.

Key Data Fields:

Category	Data Points
Inventory	Stock levels, reorder points, safety stock, product ID

Sales	Units sold, revenue, channel (online/retail), region
Customer Info	Region, purchase volume, demographics
Orders	Order date, status, delivery time, fulfillment rate
Suppliers	Lead time, defect rate, cost per unit
Logistics	Shipping duration, delivery success, carrier ID
Costs	Manufacturing, logistics, overhead, return cost

🎯 Project Objectives

You are tasked with building an **interactive Tableau dashboard** that delivers:

1. **Inventory health** across warehouses and SKUs.
2. **Order performance** insights including delay hotspots.
3. **Supplier benchmarking** by defect rates and reliability.
4. **Transportation route efficiency** and cost breakdowns.
5. A unified view of **total supply chain cost structure**.

Step-by-Step Approach

◆ Step 1: Data Preparation

- Clean and normalize data types for dates, IDs, and metrics.
- Join key tables logically by foreign keys: Product ID, Order ID, Supplier ID, etc.
- Create calculated fields for:

```
Inventory Gap = Current Stock - Reorder Point
```

- **On-Time Delivery %**

```
Total Cost per Unit = Manufacturing + Logistics + Overhead
```

◆ Step 2: Dashboard Development

📌 Module 1: Inventory Status by Warehouse

- **Chart Type:** Heatmap or Bullet Chart
- **Fields:** Warehouse, Product ID, Inventory Gap
- **Goal:** Identify understocked and overstocked SKUs in real time.

✅ *Insight:* Several products in East Zone warehouse consistently drop below safety levels.

📌 Module 2: Order Fulfillment & Delivery Trends

- **Chart Type:** Line Chart + KPI cards
- **Fields:** Delivery Time, Order Status, Region
- **Goal:** Analyze fulfillment efficiency over time and by geography.

✅ *Insight:* Weekend dispatches see higher delays; Region West has repeat SLA breaches.

📌 Module 3: Supplier Performance Analysis

- **Chart Type:** Scatter Plot or Table
- **Fields:** Supplier ID, Avg. Lead Time, Defect %, Unit Cost
- **Goal:** Benchmark suppliers based on cost-efficiency and quality.

✅ *Insight:* Supplier C has the lowest cost but highest defect rate — high risk.

📌 Module 4: Shipping Efficiency by Carrier & Route

- **Chart Type:** Bar Chart / Map Visualization
- **Fields:** Carrier ID, Delivery Time, Success Rate, Fuel Cost
- **Goal:** Evaluate which routes and logistics partners offer optimal cost and service.

✅ *Insight:* Carrier B on southern route is most efficient; Carrier A incurs peak-season surcharges.

📌 Module 5: Cost Structure Breakdown

- **Chart Type:** Pie Chart + Time Series
- **Fields:** Manufacturing, Logistics, Miscellaneous, Returns
- **Goal:** Visualize total operational cost drivers and seasonal variations.

✅ *Insight:* Logistics costs spike during promotional campaigns; returns have been rising in Q4.

Key Takeaways & Strategic Insights

Category	Insight
Inventory	Overstocking in urban centers leads to tied-up capital; rural warehouses risk outages.
Orders	Fulfillment delays align with last-mile delivery constraints, especially in hilly zones.
Suppliers	Balancing cost vs. quality is key. The current sourcing model favors cost, not reliability.
Transport	Hybrid model (own fleet + 3PL) offers control but increases variability in performance.

Costs	Non-optimized returns and emergency shipping inflate per-unit cost margins by ~12%.
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Business Recommendations

1. Rebalance Inventory Allocation

Implement predictive restocking models using demand heatmaps and reorder automation.

2. Supplier Rationalization Strategy

Shift toward high-reliability suppliers, even at slightly higher unit costs, to reduce rework and SLA violations.

3. Delivery Optimization

Analyze failed deliveries by region and time slot. Introduce dynamic rerouting in high-risk zones.

4. End-to-End Cost Tracking

Build automated alerts for cost spikes in logistics or raw material sourcing.

5. Centralize Operations Data

Use Tableau as a single source of truth for daily operations—integrating it with real-time systems (ERP, WMS).